Paleomagnetism of the Morin Complex

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Metamorphosed (high amphibolite to granulite facies) anorthosites and leucogabbros of the Morin complex in the province of Quebec have two magnetizations often coexisting in the same specimens. The magnetization with high remanent coercive force has a mean direction of 266.2°, −76.8° (24 sites, 117 samples, α_{95} = 5.3°) with respect to the present horizontal and a corresponding pole at 42.4°S and 139.3°E. It has blocking temperatures between 550° and 650°C, remanent coercivities chiefly between 1000 and 3000 Oe, and is due to hematite with a few percent of ilmenite in solid solution. It is suggested that this high coercive force (hcf) magnetization was acquired during cooling immediately following high-grade regional metamorphism during the Grenvillian orogeny at about −1124 ± 27 m.y. The other magnetization has remanent coercivities mostly between 100 and 1000 Oe and is referred to as the medium remanent coercive force (mcf) magnetization. This magnetization has a mean direction of 114°, + 38° (13 sites, 45 samples, α_{95} = 10°) and a corresponding pole at 0°, 164°E. It has blocking temperatures mostly in the range 100°–400°C, and we suggest that it was acquired later during the last stages of regional uplift at about −1000 m.y. The poles for the hcf and mcf components of magnetization define the limits of the Grenville polar track and provide for the first time tentative estimates of its age. There is no systematic deflection of the magnetization due to anisotropy. The close agreement between the pole for the hcf magnetization and similar poles derived from the Allard Lake anorthosite and basic gneisses near Haliburton, localities spaced over 1200 km apart, indicates that the present horizontal is the proper reference plane for the analysis of these magnetizations.

The Morin complex is an intrusive mass situated in the Grenville structural province about 50 km north of Montreal. Its situation within the Grenville province and the comparison of paleomagnetic results from it and other Grenville rocks are described in the companion paper [*Irving et al.*, 1974]. The Grenville province is the youngest and most southeasterly element of the Precambrian shield of North America, and it consists largely of highly metamorphosed rocks, which have been heated to temperatures in excess of the maximum blocking temperatures of magnetite (580°C) and hematite (675°C), so that any earlier magnetic record was erased. Hence the problems are different from those in conventional paleomagnetic work, in which rocks are assumed to have acquired their stable magnetization at or soon after deposition. Instead, the magnetization was acquired sometime after regional metamorphism. Hence there are serious problems in estimating the age of the magnetization. Further problems are the possible effects of anisotropy and the difficulties in assessing relative block rotation within such geologically complex terrain and hence in knowing the horizontal plane at the time that the magnetization was acquired. Although several pioneer studies have been made [*Hood*, 1958; *Du Bois*, 1962; *Hargraves and Burt*, 1967], paleomagnetic work on metamorphic rocks is still in its infancy. In this paper we present a study of the magnetization directions and ranges of blocking temperatures and remanent coercive forces present in a large metamorphosed igneous complex as a contribution to this youthful and rapidly expanding field.

Results are reported from 32 sites. Generally, at each site, five cores were drilled. Two specimens were cut from each core. Thermal demagnetization and alternating field (af) demagnetization were carried out by using the apparatus described by *Roy et al.* [1969, 1971, 1973] in which the ambient field was canceled to less than 2 γ.

GEOLOGY

The Morin complex was intruded partly into metasedimentary rocks, which most authors correlate with the Grenville group, and partly into gneisses that may predate the Grenville group. The complex has undergone regional metamorphism to high amphibolite or granulite facies. For convenience an igneous terminology is used. Anorthosite and leucogabbro comprise the largest unit (Figure 1). The degree of recrystallization is variable. Igneous textures are preserved in much of the western lobe, whereas the southern half of the eastern lobe is completely recrystallized. In the northern part of the eastern lobe and the east margin of the western lobe the rocks show intermediate degrees of recrystallization. Most samples are from the western lobe (24 sites). Two sites (46 and 47) are in and near the eastern lobe.

Much of the anorthosite is surrounded by an inner pyroxene diorite and pyroxene monzodiorite unit and an outer pyroxene quartz monzonite unit. The inner unit is commonly oxide rich and usually shows intrusive relations toward and contains inclusions of anorthosite. It was sampled at two sites (30 and 73). Dikelike bodies, probably related to the inner unit, occur in the anorthosite, and one was sampled (33). The outer marginal unit is transitional from the inner one. The transition occurs by a decrease in the amounts of opaque oxides and mafic silicates and by an increase in quartz and alkali feldspar. The spatial relationship of the inner and outer units suggests that the former may be the basal facies of the outer unit, which has been enriched in oxides and mafic silicates. The outer unit was sampled at three sites (70, 71, and 72).

The Morin complex underwent severe regional metamorphism and deformation after crystallization. It may have been emplaced and then deformed and recrystallized some considerable time later during the Grenvillian orogeny [*Emslie*, 1970; *Wynne-Edwards*, 1972]. Alternatively, emplacement, deformation, and metamorphism may all have occurred at approximately the same time within the Grenvillian orogeny, as was discussed by *Martignole and Schrijver* [1971]. These authors suggest that the deformation was caused by diapiric rise of anorthositic rocks into hot country rocks.

The probable age sequence of the Morin rocks from oldest to youngest is anorthosite-leucogabbro, pyroxene monzodiorite, and pyroxene quartz monzonite. There are no

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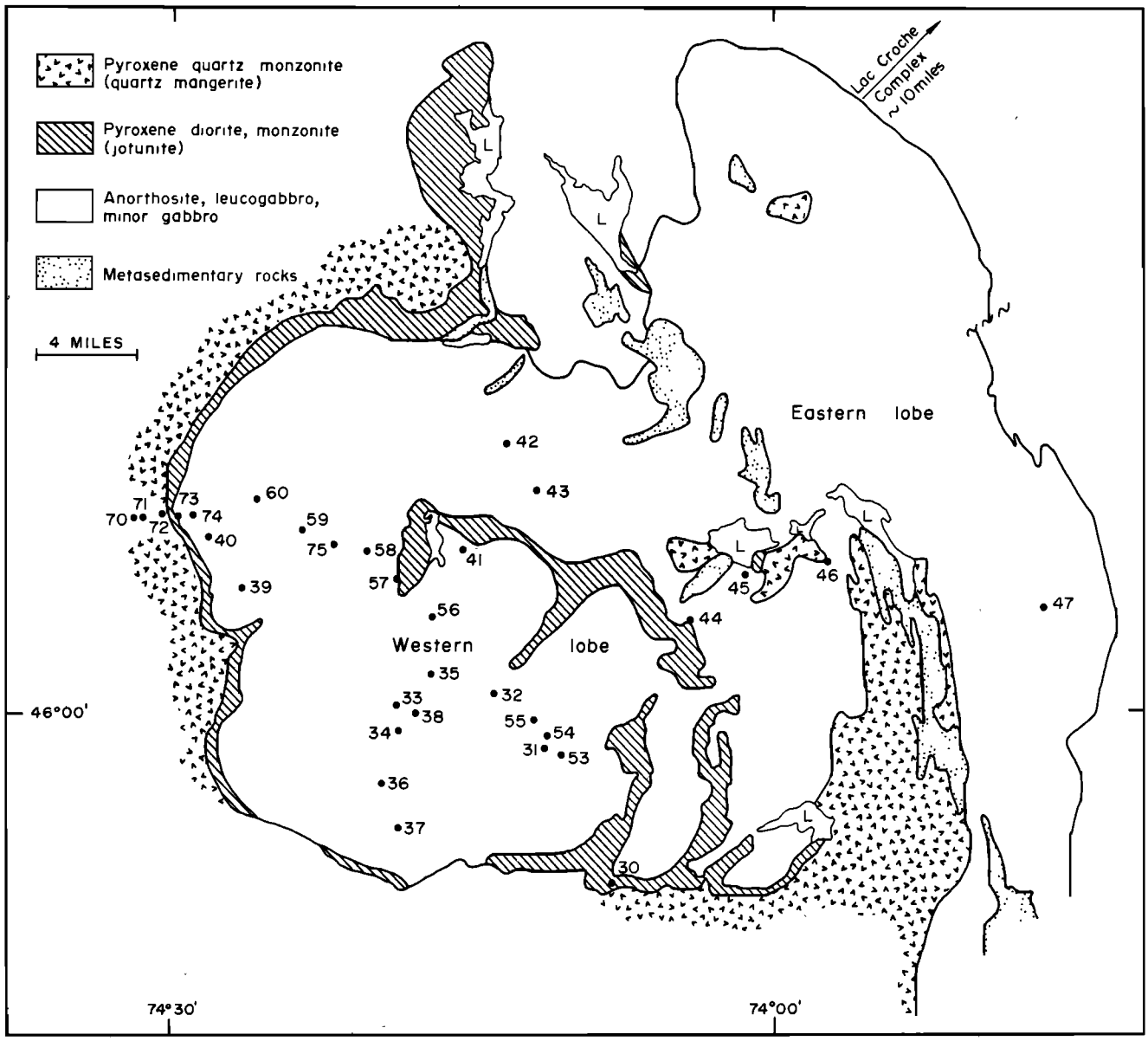


Fig. 1. Sketch map of the Morin complex. Geology is from reports of the Quebec Department of Natural Resources and R. F. Emslie (unpublished data, 1974). Sampling sites have laboratory accession numbers that have no geological significance. Site 75 is the magnetite ore body at Desgrosbois, results from which are not reported herein.

published age determinations from the Morin complex itself, but the date of its high-grade regional metamorphism can be inferred from other data (see below). The K-Ar biotite and hornblende ages from southwest Quebec range from −910 to −1085 m.y. Whole rock Rb-Sr isochrons from the Lac Croche complex about 12 km northeast of Morin have been interpreted as indicating that intrusion, deformation, and metamorphism took place about −1124 ± 27 m.y. [*Barton and Doig*, 1972]. The Lac Croche complex comprises rocks of the anorthosite suite and would be expected to be similar in age to the Morin. *Barton and Doig* [1973] have reported a Rb-Sr isochron of −1094 ± 43 m.y. from rocks that they consider to belong to the Grenville group, near the eastern margin of the Morin complex, and they interpret this to be the age of regional metamorphism. Barton and Doig have also obtained a Rb-Sr isochron of −1576 ± 19 m.y. from the lower part of the Lac Quinn formation, which is close to the margins of the Morin complex [*Martignole and Schrijver*, 1970]. They suggest that this older isochron has survived because of dehydration during an earlier metamorphism. Alternatively, the old isochron from the Lac Quinn formation can be interpreted as reflecting the contact metamorphism caused by initial intrusion of the Morin complex, which would therefore be of this age. It is noteworthy that about 170 km to the east, *Frith and Doig* [1973] have reported an isochron of −1482 ± 72 m.y. from rocks believed to be in the thermal aureole of the Lac St. Jean anorthosite. There seems to be no reason to doubt that the −1094- and −1124-m.y. isochrons date the high-grade regional metamorphism of the Morin and Lac Croche areas. The younger K-Ar ages may reflect uplift following regional metamorphism or a later mild thermal event.

HCF AND MCF MAGNETIZATION

Demagnetization results are illustrated in Figures 2, 3, and 4. At site 32 the natural remanent magnetization (NRM) direction is toward the east and down. During af demagnetization

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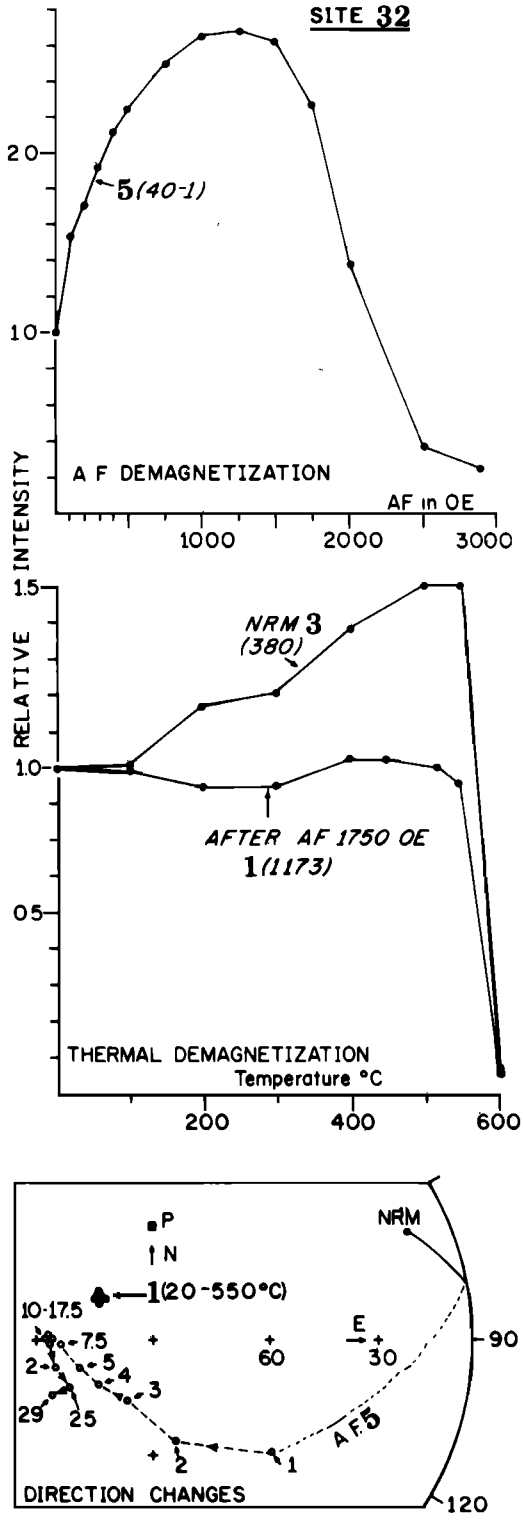


Fig. 2. Demagnetization of anorthosite site 32. Relative intensity is plotted against peak af (top) and temperature (middle). The intensity of NRM (microgauss) is given in parentheses. Also shown are the corresponding variations in directions (bottom). *P* is the present field (downward), and the open circles denote negative inclinations. The points are labeled in 10^2 Oe (top) and 10^2 °C (middle). The large numbers are specimen numbers.

there is a large increase in intensity, and the directions migrate to the west with upward inclination (Figure 2, bottom) with a stable end point between 750 and 1750 Oe. The directions do not fall on the shortest route between the end point and the present field. Hence the magnetization removed during de-

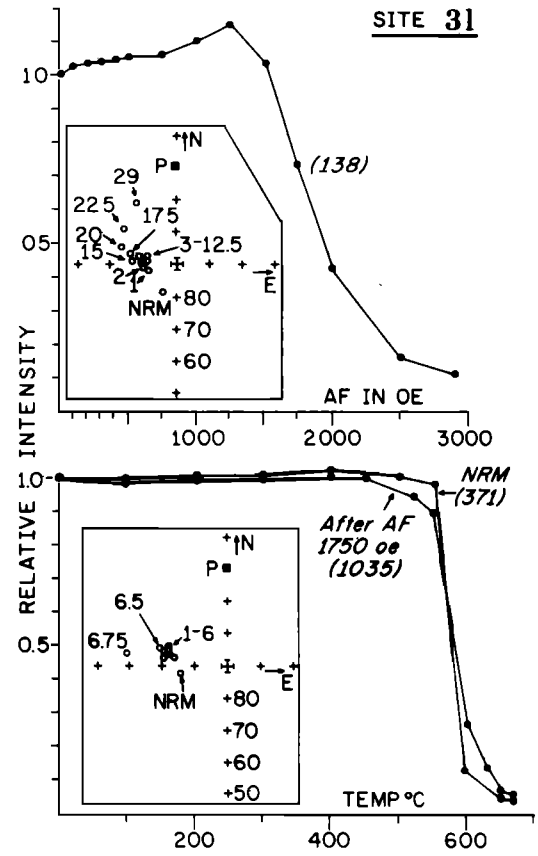


Fig. 3. Demagnetization of anorthosite site 31. Open (solid) circles denote upward (downward) inclinations. The corresponding directional variations are given in the insets. Otherwise, legend is same as that for Figure 2.

magnetization cannot have been imposed by the present field. In Table 1 the magnetizations measured at each demagnetization step up to 2000 Oe are listed, together with the magnetization removed at each step calculated by vector subtraction using first the NRM and then the 100-Oe step as a basis.

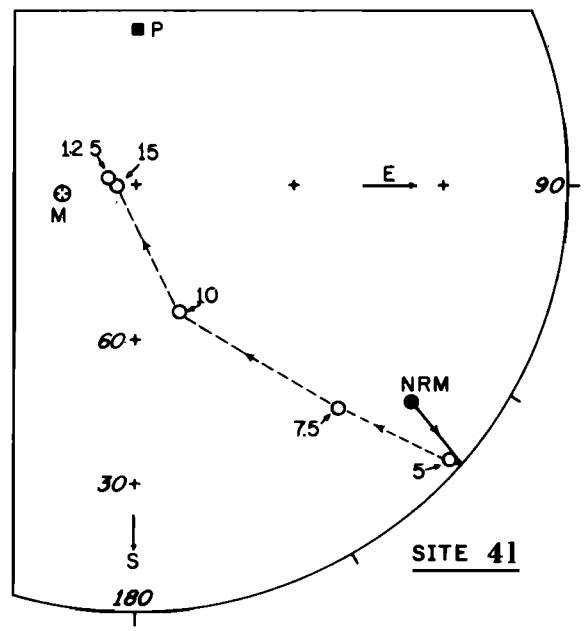


Fig. 4. Changes in direction during af demagnetization of anorthosite specimen from site 41; *M* is the mean direction for all anorthosite rocks (Table 4).

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TABLE 1. Vector Analysis of af Demagnetization of Specimen From Site 32

af	Measured		Difference 1		Difference 2	
	M	*D*, *I*	*M*	*D*, *I*	*M*	*D*, *I*
NRM	40	057, +13	...	...	...	...
100	63	134, -47	75	002, +47	...	...
200	67	167, -63	89	027, +50	28	090, +30
300	77	204, -73	102	045, +54	50	103, +35
400	86	230, -72	112	054, +54	64	104, +34
500	91	249, -70	118	062, +54	74	107, +33
750	101	267, -66	127	072, +53	90	111, +31
1000	106	270, -63	133	075, +51	97	111, +30
1250	107	270, -64	134	075, +52	97	111, +31
1500	105	272, -63	132	076, +51	97	112, +30
1750	92	266, -64	119	072, +50	84	111, +26
2000	56	254, -64	86	064, +43	59	113, +04

M is the intensity of magnetization in microgauss. *D* and *I* are the declination and inclination. The measured vectors are listed on the left. On the right the vector removed by demagnetization in the af in oersteds given in the first column is listed. Difference 1 is NRM minus the magnetization in a given step. In difference 2 the magnetization at 100 Oe is used as the base.

In the stable end point range (750–1500 Oe) the direction and intensity of both measured and removed magnetizations are essentially constant. The vector removed in 100 Oe is closer to the present field than later ones, the indication being that a small component related to the present field and with low remanent coercive force is present. Thus the NRM consists of three magnetizations. The first has remanent coercive forces exceeding 1500 Oe (intensity 105 μG; 270, −64) and is referred to as the high coercive force (hcf) magnetization. The second has remanent coercive forces between 750 and about 100 Oe (intensity 131 μG; 074, +52) and is referred to as the medium coercive force (mcf) magnetization. The third is a small component with remanent coercive force less than about 100 Oe directed close to the present field and is referred to as the low coercive force (lcf) magnetization. The remanent coercive force spectra of the hcf and mcf magnetizations are quite distinct, so that the vector subtraction procedure is justified.

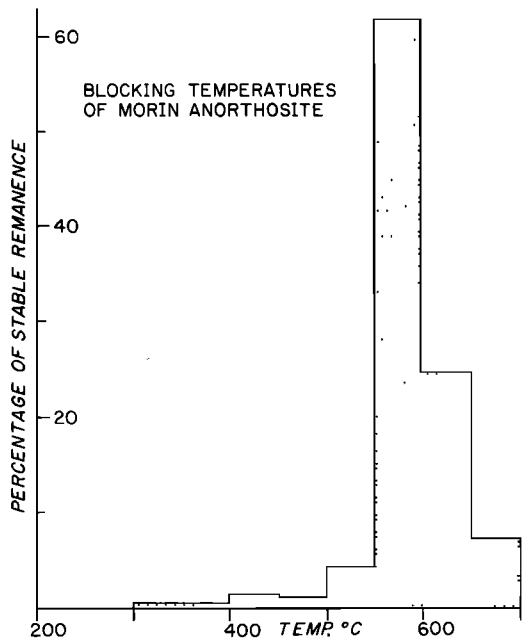


Fig. 5. Mean blocking temperatures of hcf magnetization of NRM. Result compiled from decay curves for seven anorthosite and gabbro specimens (sites 31, 32, 33, 34, 40, 42, 44) after af cleaning in the fields listed in Table 3. The blocking temperatures refer to the hcf magnetization only and not to the mcf magnetization.

During thermal demagnetization of NRM the magnetization increases up to 500°C (Figure 2, center). Thermal decay of the hcf component is very different from that of total NRM. The curve is approximately flat up to 500°C, it being shown that af cleaning at 1750 Oe successfully removes the lcf and mcf components, a useful confirmation of the af cleaning procedure. The blocking temperatures of the hcf magnetization are between 550° and 600°C. Comparison with the thermal decay curve of the NRM shows that the mcf magnetization has blocking temperatures between 100° and 500°C. The components removed between the five 100° steps have the mean direction 103, +48 (N = 5, k = 77), again directed toward the southeast and similar in direction to that removed during af demagnetization (Table 1).

At site 31 the NRM is directed upward toward the southwest (Figure 3). The direction becomes westerly after af demagnetization in fields between 300 and 2000 Oe. The remanent coercive forces are mostly between 1500 and 2500 Oe. During stepwise thermal demagnetization the directions move by a small amount to the west and remain constant between 100° and 650°C. The lower part of Figure 3 shows the decay of remanence during thermal demagnetization of the total NRM and the hcf magnetization.

Both curves are similar with blocking temperatures between 550° and 675°C. At 700°C the remanence is lost. The NRM therefore consists mainly of an hcf component directed toward the west with steep upward direction. However, there is a small mcf magnetization that is not related to the present field. Vector subtraction gives an hcf magnetization (145 μG; 280, −80) with remanent coercivities over 1500 Oe and an mcf magnetization (15 μG; 156, +39) with remanent coercivities less than 300 Oe. The remanent coercive force spectra of the mcf and hcf magnetizations are distinct, and vector subtraction is justified. There is no detectable lcf magnetization.

A further example illustrating the presence of both hcf and mcf magnetizations is given in Figure 4. Vector subtraction shows that the mcf component has a direction 129, +26 and there is no detectable lcf magnetization.

The mean blocking temperature spectrum of the hcf mag-

TABLE 2. Effect of Anisotropy

Parameter	Value
*Applied Field = 20,000 Oe**	
IRM	
mean direction (*D*, *I*)	285°, -77°
precision k	24
error α_{95}	10°
Alternating Field = 1500 Oe	
IRM	
mean direction (*D*, *I*)	381°, -73°
precision k	192
error α_{95}	4°

The applied field direction is approximately parallel to the hcf magnetization. Results are based on samples from 10 sites (N = 10). The method is explained by *Irving and Park* [1973].

*Direction is 282°, -73°.

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TABLE 3. Direction of hcf Magnetization

Site	UTMG	af	N	$D°$, $I°$ (R)
31	623917	1750	4	277, -76 (3.99)
32	609967	1750	5	289, -69 (4.96)
33an*	528947	1750	5	279, -77 (4.88)
34	528928	1000	4	245, -84 (3.93)
35	554962	2000	5	311, -69 (4.96)
36	523895	1500	5	229, -84 (4.91)
37	535865	1250	5	280, -76 (4.99)
38	539941	1750	5	293, -70 (4.97)
39	424021	1500	5	271, -66 (4.91)
40	406050	1500	5	234, -70 (4.98)
41	574039	1250	5	315, -83 (4.78)
42	623115	2000	5	280, -68 (4.98)
43	650068	1250	5	280, -64 (4.92)
44	720009	1500	5	309, -61 (4.97)
45	751035	1250	5	308, -73 (4.93)
46	806038	•••	•••	Table 8
47	932005	•••	•••	Table 8
53	638914	1500	5	213, -51 (4.90)
54	623925	1250	5	233, -81 (4.88)
55	616938	1250	5	268, -69 (4.96)
56	552005	1500	5	132, -82 (4.97)
57	525031	1500	5	298, -70 (4.71)
58	511043	500	5	142, -76 (4.79)
59	472055	1500	5	212, -75 (4.95)
60	439077	1750	5	265, -76 (4.96)
74	395063	1750	4	191, -67 (3.89)

Site numbers are given in the first column. UTMG is the universal transverse Mercator grid reference (all numbers are prefaced by the number 18). The af is the peak cleaning field in oersteds. N is the number of cores (usually two specimens from each). $D°$, $I°$ is the mean site direction giving cores unit weight. R is the resultant.
*Results are from anorthosite samples.

netizations is shown in Figure 5. They are predominantly between 550° and 650°C, and the directions are always directed to the west with steep negative inclination.

Remanence Anisotropy

A magnetic field of 20,000 Oe was applied to 10 specimens in a uniform direction (282, −73) relative to the specimens in situ (Table 2). The remanence produced is not uniform (standard deviation 16°). The mean direction is 4° from the applied field, but this departure is not significant. The specimens were then cleaned in an af of 1500 Oe. The standard deviation falls to 6°, which is the dispersion of the hcf magnetization due to anisotropy. The mean direction of the hcf of isothermal remanent magnetization (IRM) is parallel to the applied field direction. This shows that the deflections due to anisotropy are random and no systematic effects are present.

TABLE 4. Summary of Results From Morin Anorthosite

Quantity and Usual Symbol	$M1$	$M2$
Mean of sites		
Mean direction ($D°$, $I°$)	266.2°, -76.8°	114°, +38°
Number	24	13
Precision k*	32	17
Error α_{95}*	5.3°	10°
Model A pole		
Location†	42.2°S, 140.6°E	0°, 164°E
Error (dm, dp)	9.9°, 9.2°	12°, 9°
Model B pole		
Location§	42.4°S, 139.3°E	•••
Precision K	11	•••
Error A_{95}	9.1°	•••

$M1$ is the hcf magnetization. $M2$ is the magnetization removed during cleaning (mcf).
*Method of *Fisher* [1953].
†*Irving and Ward* [1964].
§*Creer et al.* [1959].

Remanence Direction of Western Lobe

Stable end point fields were determined for one specimen from each core. The other specimen from the core was then cleaned in this field, generally between 1000 and 2000 Oe. The mean directions of the hcf magnetization are given in Table 3 and summarized in Table 4. The two-tier statistical analysis yields a within-site precision k_W = 48 and a between-site precision k_B = 38. The between-site precision is significant. The within-site dispersion (standard deviation 11°) is greater than experimental errors and is about twice the scatter expected from the effect of anisotropy. The scatter could be due to secular variation during cooling. However, although small systematic changes in direction occurred during high-af demagnetization, they were not found during high-temperature demagnetization (Figure 3). We have no satisfactory explanation for the cause of this within-site scatter of the hcf magnetization.

The mean direction of the hcf magnetization has a corresponding pole at 42°S, 141°E (Table 4). This pole is in excellent agreement with poles from the Allard Lake anorthosite, eastern Quebec (39°S, 140°E [*Hargraves and Burt*, 1967]) and the basic gneisses near Haliburton, Ontario (39°S, 144°E [*Buchan and Dunlop*, 1973]). This agreement justifies the assumption that the present horizontal plane and the paleohorizontal at the time that the magnetization was acquired are coplanar.

The direction changes during demagnetization in many specimens (40%) are negligible, and only an hcf magnetization is present. Vector subtraction was carried out on all other specimens to determine the direction of the magnetization removed during cleaning. Results from 20% of the samples show a wide scatter in many cases about the present field, but the directions from 37% of the samples (from 13 sites) show a systematic grouping of the mcf components toward the southeast with downward inclination (Tables 4 and 5). The corresponding pole is at 0°, 164°E. The remaining samples (two sites, 33 (1) and 43) have southerly directions.

TABLE 5. Mean Directions of mcf Magnetization

Site	n	$D°$, $I°$ (R)
31	6	120, +46 (4.88)
32	9	093, +40 (7.91)
33an* (1)	3	110, +14 (2.91)
33an* (2)	3	192, +12 (2.90)†
34	2	138, +41 (1.90)
35	9	128, +10 (7.67)
36	8	113, +27 (6.47)
37	8	114, +30 (7.57)
39	8	061, +56 (7.04)
40	2	123, +53 (1.98)
41	9	021, +51 (7.26)
43	8	190, +54 (7.58)†
54	8	106, +45 (7.75)
55	3	122, +37 (2.75)
56	6	112, +28 (5.67)

Parameter n is the number of individual specimens, $D°$, $I°$ the mean direction, and R the resultant.
*Results are from anorthosite samples.
†Results are omitted from analysis of Table 4.

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The results from the western lobe may now be summarized. There is an hcf magnetization (coercive force 1000–3000 Oe) in all specimens with a narrow range of blocking temperatures (predominantly 550–650°C) that is directed toward the west with steep negative inclination. About one third of the specimens also contain an mcf (coercivity 100–1000 Oe) magnetization directed toward the southeast with moderate positive inclination and with lower blocking temperatures distributed over a wider range (100–500°C).

IRON-TITANIUM OXIDES OF WESTERN LOBE

Photographs of polished sections are shown in Figure 6. The presence of small quantities of impurities in solid solution cannot be detected by optical methods, and in the following description the terms 'ilmenite,' 'magnetite,' and 'hematite' are for minerals that are predominantly of these compositions. The estimated proportions of the five opaque mineral species present in the anorthosites and leucogabbros are given in Table 6. Together they comprise about 1% by volume.

All sections have large grains (100–1000 μ) with exsolution intergrowths of ilmenite and hematite (Figure 6*a*). The exsolution lamellas commonly are about 10 μ thick. Lamellas of ilmenite contain secondary exsolution intergrowths of hematite, and lamellas of hematite secondary intergrowths of ilmenite. The secondary intergrowths range in thickness from about 2 μ down below the resolving power of the microscope (about 0.5 μ). All sections contain myriads of small rods dispersed in the feldspars (Figure 6*b*). The largest are 50–1000 μ long and 2 or 3 μ thick (Figures 6*c*, 6*d*). These large rods are dominantly hematite. Far more numerous are much smaller rods, a few microns in length and less than 1 μ wide. Positive identification of these small rods was not possible optically, but they are bireflective and therefore not magnetite. Curie temperatures were determined for three anorthosite specimens (Table 7). Two have two Curie points, and the third a single value. The higher values (T_2) range from 595° to 630°C and are lower than the value of pure hematite (675°C). Thus the mineral optically identified as hematite is probably ilmenohematite with about 5–10% of ilmenite in solid solution. *Anderson* [1966] found that the average hematite content of opaque inclusions in feldspar megacrysts from the Labrieville anorthosite is always substantially higher than that in hemoilmenite of the groundmass. This is in agreement with our results, in which coarse groundmass grains are nearly equal parts of hematite and ilmenite, but at least the larger opaque rods in the silicate minerals are predominantly hematite.

Magnetite occurs as large (100-μ) discrete grains, as inclusions in ilmenite, and as grains (10–50 μ) around the margins and along the cracks in pyroxene crystals. Only rarely do ilmenite intergrowths occur. The lower Curie temperatures (T_1 of Table 7) are close to the temperature of pure magnetite (580°C), the indication being that the mineral identified optically as magnetite is essentially titanium free.

We may now ask which of these mineral phases is responsible for the hcf and mcf magnetizations. On average, about 7% (Figure 5) of the hcf magnetization has blocking temperatures between 650° and 700°C and is therefore probably caused by essentially titanium-free hematite. However, most of the hcf magnetization has blocking temperatures between 550° and 650°C and is caused either by hematite with 5–10% of ilmenite in solid solution or by that mineral plus titanium-free magnetite. The remanent coercive forces commonly exceed 2000 Oe, values that are usually associated with hematite rather than with magnetite. However, according to *Evans and McElhinny* [1969], *Evans et al.* [1968], and *Murthy et al.* [1971], rod-shaped magnetite can have such high coercivities. Rod-shaped opaque minerals are always present in anorthositic rocks of the western lobe, but they are not magnetite.

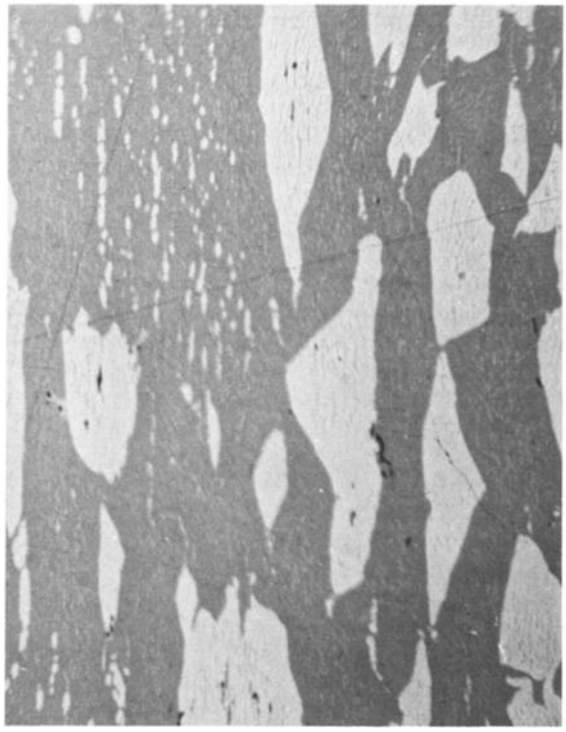


Fig. 6*a*. Polished section, site 34 (×275), exsolved hematite (light grey) and ilmenite (darker grey) within a large opaque grain; the photograph is 0.35 mm long.

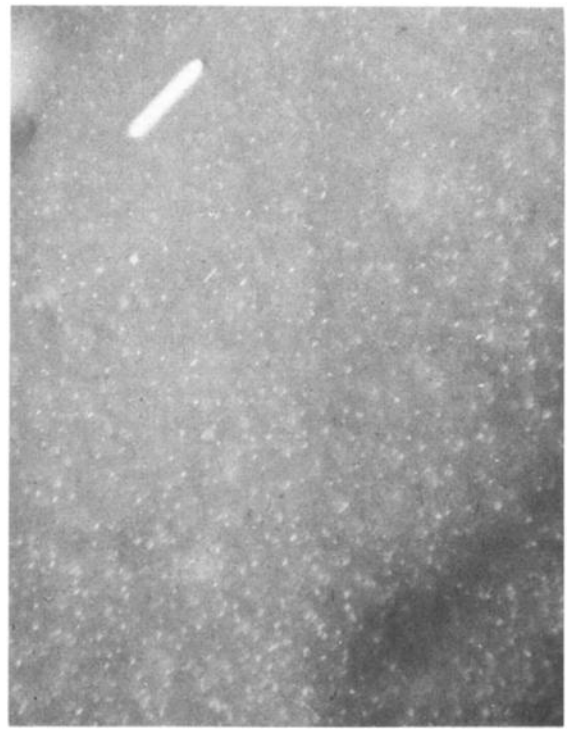


Fig. 6*b*. Polished section, site 44 (×615), myriads of small bireflective grains within plagioclase and a single much larger rod; the photograph is 0.15 mm long.

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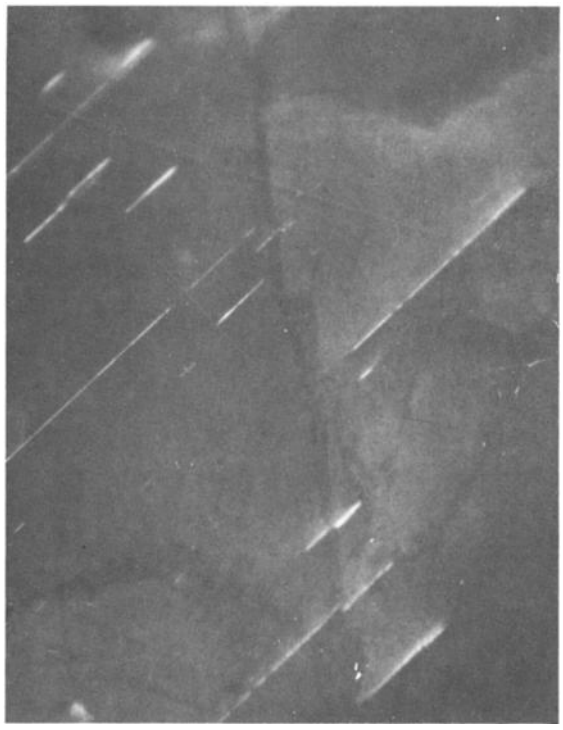


Fig. 6c. Polished section, site 32 (×615), long oriented rods in plagioclase, many of which appear to be hematite; the photograph is 0.15 mm long.

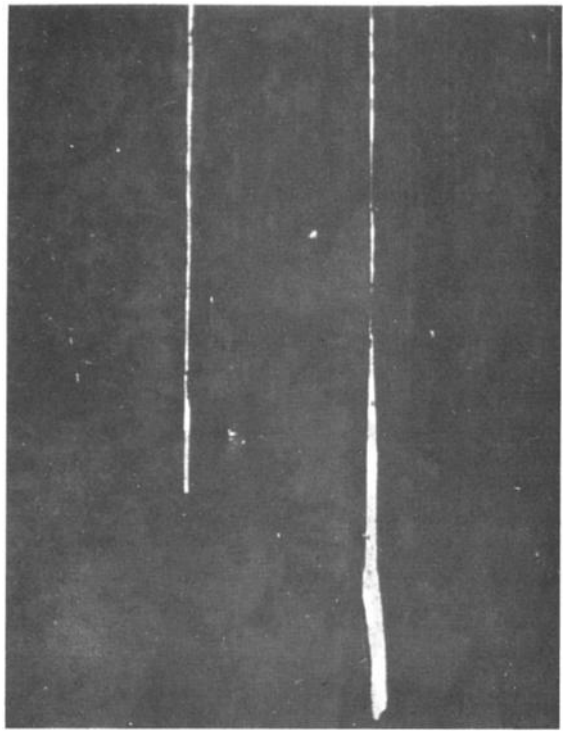


Fig. 6d. Polished section, site 34 (×128), elongate rods or plates in pyroxene, hematite, and ilmenite partly altered to leucoxene; the photograph is 0.74 mm long.

Magnetite occurs commonly in large homogeneous grains with only rare ilmenite intergrowth, a configuration usually associated with rocks with remanence of low stability [*Larson et al.*, 1969; *Wilson et al.*, 1968]. Moreover, a number of our samples with strong and stable remanence (sites 33, 34, 40, 42, 44) contain little or no magnetite. We conclude that the hcf remanence is predominantly caused by ilmenohematite with 5–10% ilmenite. *Hargraves and Fish* [1972] have reached a similar conclusion in their study of the Egersund anorthosite of Norway. The most likely source of the hcf component is the abundant very small bireflective rod-shaped crystals in the feldspars, but the small second generation hematite intergrowths in the large composite groundmass crystals also could contribute.

The mcf magnetization is probably due to magnetite, because the coercive forces and blocking temperatures are generally too low for hematite. However, hemoilmenite could be responsible.

TABLE 6. Opaque Minerals of Anorthositic Rocks of the Western Lobe

	Range, %	Mean, %
Ilmenite	20-60	46
Hematite	10-55	31
Magnetite	0-60	12
Pyrite	0-30	9
Leucoxene	0-5	2

Values are expressed as percentage of total opaque minerals.

OTHER RESULTS

A pale green sugary-textured recrystallized anorthosite at site 47 has directions that are stable and point south (Table 8), comparable to the mcf magnetization at sites 33 (2) and 43 (Table 5). This direction has not previously been reported from the Grenville province. The rock contains large ilmenite grains with hematite intergrowths like those in the western lobe. Minute bireflective grains occur disseminated throughout the rock but are less numerous than in unrecrystallized anorthosite. Magnetite was not identified.

At site 46, partially recrystallized anorthosite has westerly directions (Table 8). Large intergrown ilmenite-hematite grains occur together with minute bireflective grains whose distribution is unrelated to feldspars. Traces of magnetite are present. Curie temperatures correspond to pure magnetite and ilmenohematite (Table 7).

Rocks other than anorthosites and leucogabbros were sampled at six sites (Figure 1), but at only two (site 30 and the basic dike at site 33) were stable end points obtained (Table 8). The initial intensities are generally high and fall very rapidly in increasing af, this situation leaving an mcf magnetization that comprises only a few percent of the NRM (Figure 7). There is no detectable hcf magnetization. The thermal decay curves are

TABLE 7. Curie Temperatures in Morin Rocks

Site	T_1	T_2
30	565	675
34	•••	595
35	565	613
46	575	630

T_1 represents Curie temperatures in degrees Celsius appropriate to magnetite and titanomagnetite. T_2 represents those appropriate to ilmenohematite. Data are from G. Pullaiah (unpublished data, 1974).

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TABLE 8. Results From Eastern Limb of Morin Anorthosite, From Associated Mostly Marginal Rocks, and From Lac Croche Complex

No.	UTMG	af	N	D°, I° (R)	Pole
			Associated Rocks		
30	665838	300-350	6	274, +03 (5.68)	4°N, 166°W
33d*	528947	300	3	090, 0 (2.84)	0°, 164°W
70	359064	250-300	4	random (1.39)	...
71	364064	200-250	5	random (1.35)	...
72	380061	200	4	random (2.57)	...
73	387063	350	9	random (3.24)	...
			Anorthosite Eastern Localities		
46	806038	750	5	271, 0 (4.23)	01°N, 165°W
47	932005	1250	3	187, +26 (2.91)	30°N, 98°E
			Lac Croche Complex Leuconorite		
103	983499	400	6	242, -80 (5.58)	52°S, 136°E

Corresponding poles given in last column. Other symbols are same as those in Table 3. The low R values at sites 70-73 show them to have random directions at $P = 0.05$.
*Results are from a diorite dike.

dominated by large lcf components, and blocking temperatures are low. The blocking temperatures of the mcf component could not be determined because of the high background noise due to the large unstable components. At site 30, oxides comprise about 6% of the rock by volume. Magnetite and ilmenite occur in large grains (200 μ). Hematite is present as exsolution lamellas in ilmenite. The Curie temperature is 565°C with a small component at 675°C. The dike at site 33 contains large (100-μ) homogeneous grains of magnetite and ilmenite with hematite intergrowths.

Age and Origin of Remanence

During metamorphism to high amphibolite to granulite facies the region was heated to temperatures over 600°C for thousands, perhaps millions, of years. No earlier magnetization could have survived. The blocking temperatures of the hcf magnetization exceed 550°C. It is therefore reasonable to suppose that this magnetization dates from the time of cooling immediately following regional metamorphism. The age of metamorphism of the nearby Lac Croche complex has been estimated to be -1124 ± 27 m.y., and that of the Morin complex, with less certainty, at about the same (see above). Therefore the directions of the hcf component of the Lac Croche rocks ought to be the same as those of the Morin complex. In Table 8, results from one site from the Lac Croche complex are given. They are indistinguishable from the directions observed from Morin. Hence we regard the age of the hcf component of the Morin as being somewhat younger than -1124 ± 27 m.y.

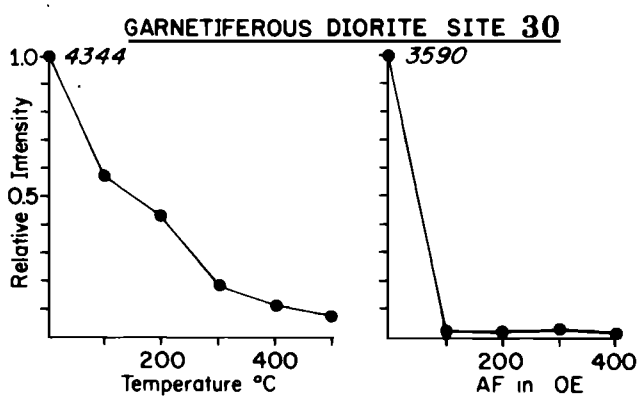


Fig. 7. Decay curves from pyroxene diorite at site 30. NRM intensities (microgauss) given.

Anderson's [1966] work on the Labrieville anorthosite suggests that the oxides exsolved from plagioclase at 800°C or more. If the Morin oxides in the plagioclases are similar, then their magnetization is probably a thermoremanence. The larger composite oxides probably equilibrated at lower temperatures, perhaps as low as 550°C. Hence their magnetization could be either a thermoremanence or a thermochemical remanence [*Nagata and Kobayashi*, 1963].

The properties of the mcf magnetization (see above) are like those of a viscous partial thermoremanent magnetization. It was probably acquired as a result of cooling during uplift. As a first approximation it is reasonable to relate the mcf magnetization to the K-Ar mineral ages obtained locally (approximately -1000 m.y.).

The marginal rocks at site 30, the dike at site 33, and partially recrystallized anorthosite at site 46 give E-W directions with reversals. The magnetizations are less stable than those of the anorthosite and are probably younger than the magnetization of the anorthositic rocks of the western lobe. The meaning of these results will not become clear until they are better surveyed.

Acknowledgments. We are very grateful indeed to Jean Hastie for much help in data reduction, to Jean Roy for his extensive help in discussion and for criticizing the manuscript, and to G. Pullaiah for permission to quote unpublished Curie point determinations. We wish to acknowledge the very useful comments of referees. Earth Physics Branch contribution 492.

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(Received January 21, 1974;
revised June 25, 1974;
accepted July 11, 1974.)

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